

KING'S COLLEGE LONDON

6CCP3131 Third Year Project

DARK MATTER COMMUNICATION
IN THE AI ERA

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Abstract

Dark matter is one of the largest open problems in modern physics, and one of the hardest to communicate to audiences without a technical background. This report documents a third-year project on communicating dark matter across five formats, each aimed at a different audience: a Substack opinion piece, a tabloid newspaper article, a podcast for first-year undergraduates, a comic for children, and a short-form advocacy video for young adults. Each piece was produced with the help of large language models, and the role AI played was deliberately varied across the module. The clearest pattern that emerged is the line between objective and subjective work: AI handled rule-bound tasks well, but struggled the more the work depended on creative judgement. A final reflection draws the experience together, including what the advocacy task revealed about the limits of accurate communication when set against simpler, more emotionally direct content.

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1 Introduction

"If I have seen further it is by standing on the shoulders of Giants."

Isaac Newton [1]

WHAT makes humans different from other animals is not intelligence by itself. Some animals are clearly intelligent. Octopuses are an example. They can solve problems and adapt to situations in ways that are surprising for an invertebrate. But none of them can communicate the way humans do, as far as we know. We can form words. We can pass an idea from one mind to another with enough precision that the next person can build on it. A child today does not have to rediscover fire. A student does not have to reinvent calculus. Most of what each generation knows has been inherited from the one before it. That is the simple point Newton was making.

This is something science especially depends on. Physics, biology, and any technical field only move forward because an idea worked out by one person can be understood and used by someone else. Communication inside science, between scientists themselves, is the layer the academic system is built around. The harder layer is communication outwards, towards people who do not have the technical background. That second layer is what this project is about.

Dark matter is a good case to think about here. The physics is technical and the research papers are written for specialists. But the basic idea is striking enough that almost anyone can find it interesting if it is presented well. About 85% of all the matter in the universe is invisible to us [2]. We know it is there because of how the rest of the universe behaves. We still do not know what it is. Once that situation is explained clearly, most people start asking questions on their own.

This project is about the problem of doing that communication well. It looks at what is actually needed to explain dark matter to audiences who do not study physics, in the kinds of formats those audiences would normally come across.

The other part of the project is the use of AI. Large language models are now part of the writing process, whether anyone planned for it or not. Science communication is difficult, so any tool that might help with it is worth looking at seriously. The same idea applies more widely. AI is changing how a lot of work is done, and the useful question is where it actually improves the output and where it does not. The point is not really about doing less work. It is more about whether the work itself becomes better when the AI is used at the right stage. None of this is obvious before trying it. The aim across the tasks that follow is to find out by working through them in practice.

The report goes through each task in order, and ends with a final reflection that brings the whole experience together.

2 Opinion Piece for a General Audience

2.1 About the Task

The first task was a Substack opinion piece on why dark matter matters, aimed at intellectually curious adults. The main source was Bertone and Hooper's 2018 review paper in *Reviews of Modern Physics* [3]. The paper traces the history of dark matter, starting from early speculation about invisible matter and going all the way to the current search for WIMPs and axions. It is a long paper. Being close enough to the physics to follow the arguments, but still far enough away to see the subject from the outside, made it easier to think about how it could be communicated. When the reader has no physics background, that distance is actually useful.

The final piece is presented in Appendix A.1.

2.2 Reflection

The Substack audience expects something personal. These are people who read science topics because they are genuinely interested, and they want to hear someone thinking through an idea, not being lectured at. That is why numbers were used as hooks. Saying that visible matter makes up only 5% of the universe [2] is a simple fact, but it makes a reader stop and think about it. The ocean analogy worked for a similar reason. When the audience does not have the physics background to follow the technical detail, analogies taken from everyday life are often the things that help.

For this task, AI (ChatGPT 5.2 Instant) was used as the primary writer from the start. There was no personal draft. A detailed prompt was written and iterated across three rounds (see Appendix B.1). The first prompt contained too much information, and the AI tried to develop all of it equally. The output spent roughly 70% of its length explaining what dark matter is, with only 30% on why it matters. The balance was wrong because the prompt gave the AI too many ideas to work with, and it could not decide which to prioritise.

In the second round the excess material was removed and more focus on significance was requested. That overcorrected. Roughly 80% argument and 20% explanation. Getting the right balance in the third round required stating the proportions explicitly in the prompt: "40% explanation, 60% argument." That is not something that would be needed when writing it yourself, but with AI it turned out to be necessary. A group review later pointed out that paragraphs 6, 7, 9, and 11 all said the same thing, that physics is incomplete. This had not been noticed during writing because the focus was on the curiosity-driven part of the argument rather than on the repetition.

The other problem was tone. The prompt included phrases from the example readings, things like "I have been thinking about this a lot lately," to push the output toward a personal voice. The AI used them, but the text still did not read like a real Substack post. The phrases were there, but the personality behind them was not.

During the group review, colleagues also noticed that many paragraphs had very similar lengths. That is another pattern that gives away AI writing. In a format where the whole point is that a real person is sharing their thoughts, that kind of uniformity works against the piece.

Looking back, asking AI to write the entire piece from scratch was not the right approach. A different order seemed worth trying: write the draft first, then bring AI in afterwards. Task 2 was where that was tested.

3 Newspaper Article for a Tabloid Audience

3.1 About the Task

Task 2 was a newspaper article on the S1 dark matter stream, written in shock-journalism style for a general newspaper audience. Short paragraphs, emotional language, and attention-grabbing claims. The source was a 2018 paper in Physical Review D, which studied how dark matter detectors could be used to measure the S1 stream [4]. The paper suited this format well. The S1 stream was referred to by the authors themselves as a “dark matter hurricane,” [4] which is already dramatic enough for a tabloid headline on its own. The science behind it is that the stream is what remains of a dwarf galaxy that was absorbed by the Milky Way around nine billion years ago. So there are large numbers and a vivid story to work with. That is exactly what shock journalism requires.

The final article is presented in Appendix A.2.

3.2 Reflection

Task 1 ended with a clear conclusion: the narrative needed to come from the writer, and AI should only come in afterwards to reshape what had already been written. That is what was tried here. The paper was read, the core ideas were pulled out, and a full draft was written before AI was brought in.

One thing that changed during drafting was the speed figure. The paper gives velocities in kilometres per second, which does not sound dramatic. But converting to kilometres per hour gives 1.8 million, and that number works much better in a headline. It is a small choice, but for shock journalism it matters.

For this task, ChatGPT was used specifically for the style aspect. The example tabloid articles from the assigned reading were fed into ChatGPT and it was asked to break down how the style works. From that breakdown the style section of the prompt was built, which included things like vivid dynamic verbs, emotional adjectives used sparingly, varied paragraph length, and frequent single-sentence paragraphs (see Appendix B.2).

The lesson from Task 1 played a role here too. In the previous task, the group noticed that AI-generated text tends to produce paragraphs of similar length, which makes it obvious. So this time ChatGPT was told explicitly to vary paragraph length and use single-sentence paragraphs often. That was supposed to fix one problem. It ended up creating a different one.

The output felt punchy and rhythmic, and it seemed good at first. But at the group meeting, people said it was not a newspaper article. Lines like “A blur. A scatter. A nightmare to measure” read more like literary prose, or spoken-word poetry, than tabloid science journalism.

The problem was the prompt. It was too long. Too many instructions were pulling in different directions. The first line said the target was “shock journalism (tabloid-style science reporting),” which should have been the dominant instruction. But later in the same prompt, pacing rules that effectively contradicted it were added. From what has been observed, AI does not resolve that kind of conflict. It does not decide which instruction is more important. It tries to do everything at once, and the output ends up being none of them properly. The pacing rules

were mechanically easy to follow, so the AI leaned into those. The newspaper format, which needs a more structural understanding of how real articles work, just got lost.

After the group meeting, a new prompt with a clear hierarchy was written. The primary instruction said: the output must read as a real newspaper article, above everything else. Secondary instructions only applied where they did not conflict with the primary one. The AI was also told not to write in a literary or poetic style, and not to use fragments or single-word sentences as standalone paragraphs (see Appendix B.2).

This was tested on both ChatGPT 5.2 Instant and Claude Sonnet 4.6, using the same draft as input. ChatGPT's output improved a bit, but it still had the same tendency. The tone came across more like a magazine feature than a tabloid (see Appendix B.2). Claude's output was noticeably different. It opened with a capitalised, punchy headline. Paragraphs were short but they did not turn into poetry. Quotes were attributed the way a real tabloid would do it. It read like something that could actually appear on a news website, which is what the task was asking for (see Appendix B.2).

Two main lessons came out of this task. The first was about prompt structure: telling AI what is wanted is not enough. It needs to be told what matters most. A long prompt with lots of instructions is not necessarily better. If everything is given at the same level, the AI cannot prioritise, it just does whatever is easiest. Format should come first, stated clearly. Style instructions should be marked as secondary. That is what was not done in the first prompt, and that is why the output failed. The second lesson was that not every AI model behaves the same way. Even with the Substack format in Task 1, ChatGPT had not been fully convincing. Here, with the tabloid format, the gap was bigger. The same draft and the same revised prompt produced very different outputs from the two models. Treating "AI" as a single thing misses an important variable: when the task requires matching a specific style, the choice of model is part of the methodology.

4 Podcast for Undergraduate Students

4.1 About the Task

The third task was a podcast script on dark matter detection using atom interferometers, aimed at first-year physics undergraduates. The main source was the TVLBAI Workshop Summary [5], a collaboration document written by over 200 scientists. It covers the planning of next-generation atom interferometer experiments, including the UK-based AION project [6].

The range of material is broad. It goes from the basic physics of how atoms can be used as interferometers to the engineering challenges of building kilometre-scale detectors. The paper is advanced, but the core ideas needed for a first-year audience are quite visual. An atom wavefunction is split by laser pulses, the two paths are recombined, and the interference pattern is read out. Because these steps can be pictured, they are easier to explain in a spoken format.

The final script is presented in Appendix A.3.

4.2 Reflection

By this point a clear pattern had emerged in how AI handled different parts of the writing process. The more the work depended on style and tone, the worse it did. So for Task 3, the decision was made to write everything from scratch and only use AI where subjectivity was not involved.

The paper was read and the core ideas that would need to be explained for a first-year audience were pulled out. The explanations had to be pitched at a level where de Broglie waves [7] and optical interferometers could be used as anchoring concepts. These are things a first-year student would recognise from A-levels.

A two-host dialogue format was chosen. One student has attended a talk on atom interferometry. The other has missed it. The explanation then happens naturally through questions and answers, and the audience learns alongside the character who was not there. First-year undergraduates are essentially A-level students in their first weeks. They would ask the same kinds of questions that the listening character asks in the script.

Once the script was written, it was given to ChatGPT (5.2 Instant) with a single prompt. The AI tool that would be used to generate the audio later in the process required a specific format: character names in capitals, square-bracket performance directions, and short punchy exchanges. Without those, it could not produce a usable voice output. Example directions the AI could draw from were also listed (see Appendix B.3). The job was entirely objective. The task was to reformat the draft and add performance directions throughout. ChatGPT was chosen deliberately. From Tasks 1 and 2, it was clear that it struggles with stylistic tasks. But here there was nothing subjective. The AI just had to look at each line, pick a fitting direction, and restructure. It did this well. The directions felt natural and no changes were needed.

The script was then converted into audio using Wondercraft AI. That is where the problems appeared. The AI-generated voices were too similar to each other. By the middle of the podcast the two speakers sounded almost the same, and at certain points it was genuinely hard to tell who was talking. Classmates who chose more obviously different pairings, a male and

a female speaker, or a speaker with a German accent, got noticeably better voice separation. When working with AI audio tools, choosing sufficiently different voices matters more than it seems during the scripting stage.

The other problem is that AI-generated audio, at least right now, cannot be treated as a finished product. The voices shifted unpredictably. The pacing was uneven. It sounded rough compared to any real podcast. But that does not make it useless. If a podcast were being produced with human speakers, the script could be run through Wondercraft first. It would give a rough sense of how the dialogue flows and where the pacing drags, before anyone steps into a recording booth. It works as a drafting tool for audio, the same way AI worked as a formatting tool for the script. Knowing which stage of the process the tool belongs in is what matters.

5 Comic for Children

5.1 About the Task

Task 4 was a piece communicating the idea of wave dark matter to 10–13 year olds. The main reading was Lam Hui's 2021 review paper on wave dark matter [8], which covers the theoretical basis for why sufficiently light dark matter particles would overlap and behave collectively as a wave rather than as individual particles. The central idea to communicate was that invisible dark matter waves pass through everything, all the time. Stating that idea is simple. Getting it into the mind of a twelve-year-old who does not know what dark matter is in the first place is considerably harder. The piece had to introduce the concept and its wave nature simultaneously, in a format with almost no room for extended explanation. A comic was chosen because the format is mostly pictures, which works better for a young audience than long passages of text.

The final comic is presented in Appendix A.4.

5.2 Reflection

The comparison between ChatGPT and Claude in Task 2 had left a clear impression: Claude handled writing tasks more naturally, particularly when a specific style or register was involved. In Task 2, Claude had been brought in as a second test, after most of the work was already done. This task was the first chance to use Claude as the primary writer from the start. Claude Sonnet 4.6 was used for the script and ChatGPT was kept for image generation later, since its image tools are stronger for that kind of work.

The task required communicating the wave nature of dark matter specifically, not just dark matter in general. The solution was to avoid treating those as two separate ideas. Instead of first explaining dark matter and then introducing the wave concept on top, the whole explanation was built around a method that presents dark matter as wavelike from the start, so that by the time the reader finishes, they have absorbed both ideas without needing them separated.

The narrative given to Claude was original (see Appendix B.4). The core analogy was Wi-Fi: Wi-Fi waves are invisible, but when the router broke at home, the kid's five-year-old brother Tommy had a meltdown because his tablet stopped working. A five-year-old screaming over a broken tablet is the kind of visual gag that children immediately understand and find funny, which keeps them engaged in what is otherwise an explanation. And because Wi-Fi is itself a wave, the analogy flows naturally into the idea that dark matter might also behave like waves, enormous ones stretching across galaxies.

The statistic was also reframed. Dark matter is usually given as roughly 27% of the universe, but framing it the other way round, that 85% of all matter in the universe is invisible, is far more striking. This was the same instinct that led to converting the S1 stream's speed into kilometres per hour in Task 2.

Claude's first response was solid (see Appendix B.4). It picked up the narrative, structured it into panels with square-bracket scene directions, and kept the dialogue age-appropriate. But a few things needed changing. The opening banner read "The Universe: More Than Meets the Eye," which felt vague and unlikely to spark curiosity in a real child. Claude was asked to

replace it with something more direct (see Appendix B.4). A section where the father explained galaxy rotation curves was also stripped out, because that level of detail was unnecessary for the audience and took up panels that could not be afforded. In a comic, every panel is real estate. The final version used nine panels on one A4 page, which meant roughly nine exchanges of dialogue to carry the entire explanation (see Appendix B.4).

Once the script was finalised, the process moved to ChatGPT for image generation. Several iterations were needed to get usable outputs. The group meeting exposed the limitations clearly. The comic looked polished, but it did not look like a comic. The images were too clean, too perfect. Real comics have a rawness to them, exaggerated expressions, dynamic angles, rough lines. The AI-generated panels had none of that. The characters stood in the same poses across panels, and the backgrounds were nearly identical throughout. Facial expressions barely changed from one scene to the next, which is a serious problem in a format that relies on visual emotion to carry the story. Another issue came up across the whole group: everyone who used AI image generation ended up with characters that looked strikingly similar, regardless of the story.

The speed was undeniable. The entire comic was produced in a single day, script and images included. Without AI, a comic of this quality would take days of drawing and layout work. But speed came at the cost of personality. This connects to something that has been building across all four tasks. AI handles objective, rule-based work well: reformatting a podcast script, compressing dialogue to fit a panel count. But the more a task depends on subjective, creative, or stylistic judgement, the more the output feels generic. A comic is an art form. The personality of the artist is supposed to come through in the linework, the framing, the exaggeration. AI cannot replicate that, at least not yet.

6 Video Advocacy for Young Adults

6.1 About the Task

The fifth task was a group project: a three-minute explainer video aimed at 18 to 25 year olds on the ongoing STFC funding cuts to curiosity-driven research [9]. Five newspaper articles served as the source material, drawn from the Financial Times [10], The Guardian [11], The Times [12], Nature [13], and Times Higher Education [14]. Each reported on the same story, but from a different perspective. The video had to explain what was happening, why it mattered, and then argue a clear position. This was the first task in the module where advocacy was expected rather than neutral explanation. The decision was made to argue in favour of the cuts. The reasoning is discussed in the reflection below.

The final script is presented in Appendix A.5.

6.2 Reflection

This task brought together communication techniques developed across the earlier tasks: numbers used as hooks, analogies grounded in everyday experience, and simple emotional framing. Every previous piece had been written in favour of science. Task 5 asked what would happen if the same techniques were turned in the opposite direction.

The decision to argue in favour of the funding cuts was deliberate. The aim was to choose the more emotionally compelling side, even though the argument itself was incomplete and in places misleading. Telling someone their tax money could go to hospitals instead of dark matter research requires almost no context. The emotional logic does most of the work on its own. It also offered a way to test whether the communication tools developed over the module would work just as well when applied against science rather than for it.

ChatGPT was used to generate a rough script from the group's ideas (see Appendix B.5) and to suggest analogies. The argument itself was developed by the group. Short-form video has its own structure: the first few seconds need a hook, and the pacing has to hold attention throughout. ChatGPT helped translate what needed to be said into that format. The £1 coin analogy is worth pulling out because it complicates the broader pattern about subjective work. Earlier tasks had shown ChatGPT struggling with subjective judgement, but the £1 framing came out of asking ChatGPT for a long list of possible analogies, around twenty in total, with that one standing out from the rest (see Appendix B.5). So even in subjective territory, AI can produce something useful, as long as it is treated as a brainstorming aid rather than a final answer. One good idea out of twenty is still one good idea.

Because this was a video rather than a written piece, the delivery mattered more than in any previous task. The script was spoken by one of the group members, and the difference between the written version and the spoken version was obvious. Sentences that read well on paper were too stiff when said aloud, so they were loosened in delivery. Stock footage was edited to match the narration, and a small number of original animations were produced by the group.

The result was the most effective piece produced in the module. Some people who watched the video agreed with the argument. Some said it felt uncomfortable. Not because the video

was poorly made, but because they had not expected to be convinced by a position they would normally reject.

Looking back, the reason it worked was not that the argument was more rigorous. It was not. The argument was incomplete in places, and parts of it were misleading. It worked because the emotional logic was already there from the start. The viewer did not need to first understand what dark matter is, or why fundamental physics matters. They just needed to care about their own tax money. That is a shorter path than any of the previous four tasks had to take.

This was the unexpected part. At the start of the module the assumption had been that the communication tools were roughly neutral, in the sense that they would work for any argument as long as the argument itself was sound. That turned out to be wrong. The tools work for incomplete arguments too. In some cases they work better, because incomplete arguments are simpler, and simpler things are easier to feel.

On social media this happens often. The content that spreads is usually the content that simplifies the most, and simplification can become distortion. What science communicators are actually up against is not just other explanations, a lot of it is content where the point is engagement rather than accuracy, so accuracy gets used when it helps and dropped when it does not. Building one of those pieces from the inside, rather than just looking at examples of them, is something the previous four tasks could not have prepared anyone for.

7 Final Reflection

The clearest pattern across the five tasks is the line between objective and subjective work. AI handled the objective parts well. Reformatting the podcast script in Task 3, helping shape group ideas into the structure of a short-form video in Task 5: these were jobs where there was either a clear right answer or a clear set of rules to follow. The subjective parts went the other way. Personal voice in Task 1, tabloid register in Task 2, the visual feel of a comic in Task 4. In each case the AI produced something that was technically correct but had no personality behind it. The further a task moved from rule-following toward creative judgement, the worse the outputs became.

Where AI sat in the writing process also changed across the module. In Task 1, ChatGPT was the primary writer from the start, and that did not work. By Task 2 the order had been reversed: the article was written first, and AI was brought in only afterwards to restyle it. By Task 3 the role had narrowed further, to formatting only. By Task 5 it was helping shape a draft from ideas the group had already worked out. AI was being used later and later, and more narrowly each time. Which model was doing the work also turned out to matter, since ChatGPT and Claude did not produce equivalent outputs even on the same task.

Task 5 added something the earlier tasks could not have shown on their own. The same techniques used to communicate science accurately worked just as well, and in some cases better, when used to argue against curiosity-driven research. Accurate communication does not have the same freedom. It has to carry the full picture, which is almost always more complicated and harder to feel. Science communicators are not just competing with other explanations, they are competing with content where the goal is engagement, and accuracy is something that gets used when it helps and dropped when it does not. The only way to keep up is to make accurate work as engaging as the alternative, because being right is not enough on its own. That is also part of why the AI question matters. Anything that helps make accurate work sharper or faster is a tool worth taking seriously, because the side competing against it is already using every tool it can.

What the module showed is that AI does not replace the work, it can sharpen specific stages of it: late, narrowly, and only where the work is rule-bound, with the model chosen to fit the task. Anywhere else, the writer still has to do the work.

A Task Deliverables

A.1 Task 1 — Substack Post

Why Dark Matter Matters

I have been thinking about this a lot lately, and I keep coming back to the same feeling: dark matter sits in a strange sweet spot between the known and the unknown. We know enough about it to be confident it is real, unavoidable, and central to the universe. But we know so little about its nature that it constantly pulls us forward, asking for deeper questions. That balance is exactly why dark matter matters.

At its most basic level, dark matter enters physics because the universe refuses to behave the way visible matter alone would allow. When astronomers first measured how fast stars orbit around the centres of galaxies, the results were unsettling. The outer stars were moving far too quickly. According to everything we know about gravity, galaxies should tear themselves apart. They do not. Instead, they remain bound, rotating as if embedded in massive halos of unseen matter. This was not a one-off anomaly. The same mismatch appeared in galaxy clusters, where galaxies move as if much more mass is present than telescopes can detect.

Then there is gravitational lensing. Light from distant galaxies bends as it passes massive objects, exactly as general relativity predicts. But again, the amount of bending consistently exceeds what visible matter can produce. Something invisible is shaping spacetime. Add to this the cosmic microwave background, the afterglow of the early universe, whose tiny temperature fluctuations encode a precise inventory of cosmic ingredients. Those patterns point to a universe dominated not by atoms, but by something cold, non-luminous, and gravitationally active. Across completely independent observations, the same conclusion keeps emerging. There is matter out there that does not shine.

That is the conceptual core of dark matter. It is not defined by what it is, but by what it does. It gravitates. It clusters. It shapes the large-scale structure of the universe. Without it, galaxies would not form the way they do. The universe would look radically different. Dark matter is the silent architecture behind everything we see.

But this is where explanation ends and significance begins. Because once you accept that dark matter is real, you are forced to confront what that fact implies. Ordinary matter makes up about five per cent of the universe. Dark matter accounts for roughly twenty-six per cent. The rest is dark energy. This means that everything familiar, from stars to chemistry to life itself, is built from a minority component of reality. We live inside a cosmic exception. That alone should change how we think about our place in the universe.

Here is where my curiosity really kicks in. We often speak as if physics is nearing completion, as if we are polishing details rather than uncovering new foundations. And in some domains,

that confidence is justified. Particle physics has been extraordinarily successful. The discovery of the Higgs boson confirmed a framework that explains all observed particle experiments with remarkable accuracy. But dark matter sits entirely outside that framework. The particles and interactions we understand account for only a small fraction of what actually exists.

This is not a minor gap. It is a profound one. It tells us that a large portion of physical reality operates according to rules we have not yet uncovered. Dark matter is not just missing content. It is missing structure. And that is where the real excitement lies.

I cannot help but wonder whether dark matter is far richer than we currently imagine. We often picture it as a single particle species drifting invisibly through space. But that picture may be far too simple. Ordinary matter is governed by layers of symmetry, interactions, and mathematical organisation. There is no reason to assume dark matter is exempt from this kind of internal logic. It could possess its own organising principles, its own hidden interactions, perhaps even its own forms of complexity. A shadow sector of nature, structured but unseen.

If that is even partly true, then discovering dark matter would not just fill in a missing number in the cosmic ledger. It would open an entirely new chapter of physics. It would reveal that the universe has been running on deeper rules all along, rules that quietly shaped galaxies and cosmic history while remaining hidden from our experiments. That possibility is hard to overstate. It suggests that our current theories are not wrong, but incomplete in a way that is intellectually fertile.

The ocean analogy keeps resurfacing in my mind. We have become experts at studying the surface of reality. We can describe waves with astonishing precision. But dark matter hints that beneath that surface lies a vast depth we have barely touched. Entire layers of structure may exist below, influencing everything above without announcing themselves directly. Once you see it that way, ignoring dark matter feels almost unnatural.

This is why dark matter matters to me. Not just because it explains galaxies, but because it keeps physics honest. It reminds us that success does not mean finality. Every time science has believed it was close to the end of the story, curiosity has pushed it further. We never knew what lay on the other side of those questions. We only knew that asking them mattered.

If early humans had never wondered what would happen when two stones struck each other, fire would have remained hidden. The paradox of curiosity is that we cannot predict what it will give us. But history teaches us something reliable. Whatever lies beyond our questions is always transformative. Dark matter feels like one of those questions. And that is why I think it matters so much.

A.2 Task 2 — Newspaper Article

SCIENTISTS STUNNED as “Invisible Hurricane” Races Through Earth at 1.8 Million km/h

An invisible hurricane is screaming through you right now.

At nearly 1.8 million kilometres per hour.

And you can't see it. You can't feel it. You can't stop it.

Because it's made of dark matter.

The most elusive substance in the Universe.

Dark matter doesn't glow.

It doesn't absorb light.

It almost never collides with you or your walls or the air in your lungs.

So how on Earth do we know it's there?

Gravity.

Galaxies spin too fast. Stars whip around too wildly. Something massive is holding them together, an enormous, invisible halo wrapped around every galaxy, including our own Milky Way.

And that's where the storm begins.

The culprit has a name.

S1.

S1 is a stream of stars drifting through our galactic neighbourhood, not randomly, not chaotically, but together, moving as a pack.

Like debris from a long-forgotten cosmic collision.

About nine billion years ago, a smaller galaxy, roughly 100 times less massive than the Milky Way, smashed into us.

It was swallowed whole.

But not erased.

We are, in a sense, standing inside a crime scene.

And S1 is the lingering evidence.

If that ancient galaxy carried dark matter, and scientists strongly suspect it did, then some of that dark matter is expected to still be travelling with the stellar stream.

Blasting past us.

This is the so-called “dark matter hurricane.”

But before you start imagining apocalyptic shelters or underground bunkers, relax!

This storm won't tear buildings apart.

It won't rattle your windows.

It passes straight through you, through Earth, through everything.

Dark matter barely interacts with normal matter

at all.

That's exactly why it's so maddening.

Under normal circumstances, dark matter around our galaxy moves in messy, unpredictable directions, like toddlers running across a field, each one in their own chaotic path.

A blur.

A scatter.

A nightmare to measure.

But S1 is different.

Its stars move in a known direction.

And if dark matter is flowing with it, then that invisible matter should be streaming past Earth in a specific, predictable way, like soldiers marching in formation.

That changes everything.

Because if scientists know the direction, they can tune their detectors.

They can look where the wind is strongest.

Experiments searching for dark matter, whether they're hunting for heavy particles or ultra-light fields, rely on subtle signals, tiny interactions, faint whispers of something unseen.

Most of the time, they're listening in a storm of noise.

But a directed stream?

That's a pattern.

That's a signature.

It could boost the chances of detection.

It could leave a distinctive imprint.

Of course, there's a catch.

Dark matter might still evade us.

Our technology might not yet be sensitive enough.

Or dark matter might behave in ways we haven't even imagined.

The Universe has surprised us before.

But if there is an invisible stream of matter tearing through the Solar System, a relic of an ancient galactic collision, then this may be one of our best shots at finally catching it.

An unseen hurricane.

Racing through your body at 1.8 million km/h.

Silent. Harmless.

And possibly the key to unlocking one of the greatest mysteries in modern physics.

A.3 Task 3 — Podcast Script

PODCAST: THE MISSED TALK

Dark Matter & Atom Interferometry

Duration: ~ 6 minutes

Speakers: **SAM** (male, 1st year undergraduate physics student at KCL, international student from the US)

and **ALEX** (male, 1st year undergraduate physics student at KCL, local London student)

SAM: [*Curious tone*] Wait, did you go to that physics talk last night? The one about dark matter?

ALEX: [*Excited*] Yeah! It was actually so good.

SAM: [*Slightly frustrated*] I saw it on Instagram about two hours before it started, and I couldn't make it. I thought it was going to be another "dark matter is mysterious" talk.

ALEX: [*Laugh softly*] No, no, this was different. It was about actually building experiments to detect it, with atoms.

SAM: [*Genuine surprise*] Ah, interesting. . . Atoms finding dark matter? I thought it was more about telescopes and galaxies.

ALEX: [*Confidently*] Okay, so the lecturer started with something called atom interferometry. You know how in optics you split a light beam and recombine it to look for phase shifts?

SAM: [*Thinking*] Like the one LIGO uses?

ALEX: [*Nods slightly*] Exactly, but that's for photons. Imagine doing that, but with atoms.

SAM: [*Slight pause*] Interference for atoms?

ALEX: [*Matter-of-fact*] Yes.

SAM: [*Puzzled tone*] But how does that work? How can we interfere particles? Isn't interference just a thing for waves?

ALEX: [*Patiently*] Correct, but you might've learned the idea of de Broglie in A-levels?

SAM: [*Recognition in voice*] Oh yeah, the fact that every particle is also a wave.

ALEX: [*Affirming*] Exactly. So essentially, we use the fact that matter has wave-like properties. Since every particle also behaves like a wave, we interfere those wave aspects of it.

SAM: [*Leaning in*] But how does the interference actually work?

ALEX: [*Explaining carefully*] What we actually do is use the quantum nature of these waves. We split the wavefunction so that each part travels along its own path and eventually recombines.

SAM: [*Curious*] And then we see a pattern?

ALEX: [*Enthusiastic*] Exactly. We see an interference pattern when these matter waves recombine. But those matter waves only produce a stable pattern if there's no disturbance along the path.

SAM: [*Lower voice slightly*] What would happen if there is a disturbance?

ALEX: [*Thoughtful pause*] Well, in the case of any disturbance, we would see the pattern

change.

SAM: *[Connecting the dots]* Since the talk was about dark matter, I'm assuming the disturbances would come from dark matter?

ALEX: *[Slight nod in voice]* Exactly. Since our theories suggest that dark matter interacts very weakly with ordinary matter, that interaction could potentially be detected by our atom interferometer.

SAM: *[Skeptical tone]* I've always heard about this weak interaction of dark matter. How do we know our detectors are precise enough to detect something so weak?

ALEX: *[Impressed tone]* That's actually a really fair question. The lecturer didn't just assume it works, they explained why it could work.

SAM: *[Interested]* Okay, how?

ALEX: *[Clear and steady]* Basically, atom interferometers are extremely precise. They're known to measure gravity with precision down to about 10 to the power of minus 12.

SAM: *[Whistles softly]* Oh, that's really very precise.

ALEX: *[Emphasizing]* Yeah, and that's the key point. We're not trying to see dark matter smash into atoms like billiard balls. Instead, we're trying to detect tiny shifts in the wave-like behavior of atoms caused by dark matter.

SAM: *[Impressed]* Ah, that's really cool. Have we actually built an interferometer like this yet?

ALEX: *[Explaining]* Yes, but not at the full scale they're aiming for. Right now, there are prototypes around 10 meters tall.

SAM: *[Processing]* Okay, but if they already work, doesn't that mean we can detect dark matter with them?

ALEX: *[Shakes head slightly]* Not really. They're precise, but not precise enough for the kind of dark matter signals we're looking for. To reach the required sensitivity, they need something much bigger, like 100 meters, or even a kilometer.

SAM: *[Realisation dawning]* I'm guessing making it longer somehow increases the precision?

ALEX: *[Confident]* Yes, exactly. The longer the atoms evolve between laser pulses, the more sensitive the measurement becomes. If there's a tiny disturbance, like from an ultra-light dark matter field, the effect accumulates more over a longer baseline.

SAM: *[Half-laughing]* So right now we're at 10 meters, and the dream is 1 kilometer?

ALEX: *[Excited]* Exactly. The 10-meter versions prove the concept. The kilometer-scale one is where you start reaching entirely new sensitivity levels.

SAM: *[Mock serious]* But wait... building a kilometer-scale experiment underground sounds insanely expensive. And on top of that, ultra-light dark matter is still just a hypothesis. So how do you justify funding something that big?

ALEX: *[Calmly]* Because it's not just about dark matter. This kind of detector would be useful for many other areas of physics too.

SAM: *[Curious tone]* What else does it do?

ALEX: *[Building suspense]* It detects gravitational waves.

SAM: *[Surprised]* Like LIGO?

ALEX: *[Explaining clearly]* Yes. But here's the interesting part: LIGO detects high-frequency

gravitational waves, things like stellar-mass black hole mergers. Also LISA, another detector, detects very low-frequency waves.

SAM: *[Thinking aloud]* So we already cover high and low frequencies.

ALEX: *[Affirming]* Exactly. But there's a middle frequency band that isn't well explored. Large-scale atom interferometers are naturally sensitive in that mid-frequency range.

SAM: *[Understanding]* So it fills a gap?

ALEX: *[Firmly]* Yes. That's one of the strongest scientific motivations. It could detect things other detectors could never probe.

SAM: *[Laugh softly]* Ohhh, so it's not a single-purpose dark matter machine.

ALEX: *[Smiling in voice]* Not at all. It also tests fundamental physics, things like precision tests of general relativity and probing quantum mechanics in gravitational fields.

SAM: *[Thoughtful pause]* So even if ultra-light dark matter turns out to be wrong...

ALEX: *[Confident]* The experiment would still be incredibly valuable.

SAM: *[Impressed tone]* That makes funding it a lot less risky, doesn't it?

ALEX: *[Agreeing]* Exactly. Dark matter is one exciting target, but the broader case is that we're building a new type of quantum sensor at an unprecedented scale.

SAM: *[Reflective]* So it's more like building a new telescope, which serves many things, not just chasing a single theory.

ALEX: *[Approving tone]* That's actually a really good way to put it. When you open a new observational window, you usually discover things you didn't even predict.

SAM: *[Regretful laugh]* Okay... now I really regret missing that talk.

A.4 Task 4 — Comic for Children



A.5 Task 5 — Video Script

THE £1 QUESTION

Video Advocacy Script, STFC Funding Cuts

Format: 3-minute explainer video (YouTube Short / Instagram Reel / TikTok)

Target audience: 18–25 year olds

Imagine you had just £1 to spend on scientific research.

Would you spend it randomly, or would you put it somewhere that actually has a good chance of paying off?

That's basically the argument our government is having right now, and it's why curiosity-driven research might actually get a budget cut.

So here's the idea.

Curiosity-driven research, which is basically science done just to understand the universe, might get less funding.

We're talking about things like dark matter, black holes, and particle physics.

Instead, more money would go into areas the government actually cares about right now, like AI, life sciences, and health technology.

The logic is simple: not everything gets funded equally, and the higher priority areas get more money.

Now this matters to you because this is public money, it's literally coming from your taxes.

So the idea is that your money should be used more wisely.

And here's something to think about.

Right now, dark matter research is funded at roughly the same level as health technology.

But even if we fully understood dark matter tomorrow, there's no guarantee it would actually improve your life anytime soon.

Meanwhile, breakthroughs in medical technology could mean better treatments, new medicines, and people living longer.

So think of it like a gamble.

If we keep funding curiosity-driven research at the same level, we're basically putting a lot of money on the unknown.

Yeah, you might win big, but most of the time, you probably won't.

So ask yourself: would you really put all your money on one number on a roulette wheel?

If the goal is to make scientific research more sustainable and useful for the economy, then it kind of makes sense to prioritise the research that's more likely to give results, instead of just taking a risk on something unknown.

Because at the end of the day, if you've only got £1, where you put it really matters.

Video Stills



Opening of the video: the £1 hook delivered straight to camera.



Animated visual: the £1 weighed against alternative funding priorities, in this case economic growth and national security.

B Prompts and Supplementary Material

B.1 Task 1: Prompts and Supplementary Material

Prompt 1 (ChatGPT 5.2 Instant)

Produce a 700–1000 word opinion-style text on “Why Dark Matter Matters.” Use a confident and clear tone. Do not write in a neutral or middle-ground style. Use a personal, reflective voice similar to Substack. You can use phrases like:

I have been thinking about this a lot lately... / This got me wondering... / It has been on my mind recently... / I cannot help but ask... / Here is where it gets interesting... / To understand why, we need to go back to... / To me, the answer is pretty clear... / I know this is not the fashionable take, but... / Naturally, I looked it up... / Here is what the numbers actually show...

(You don’t need to use these exact phrases, they’re just meant to give you a sense of the tone. Feel free to use similar expressions that fit naturally into the text.)

Use the following points as a guide for the structure and the main ideas:

1. Start with the basic idea: Dark matter is real and it shapes the universe, but we cannot see it directly.
2. Mention the observations: rotation curves, galaxy clusters, lensing.
3. Mention Probably Wrong Theories: Modified Newtonian Dynamics, why it breaks down.
4. Mention Probably Correct Theories: non-baryonic particle dark matter, CDM, WIMPs and axions.
5. Use the numbers as a hook: visible matter ~5%, dark matter ~26%.
6. Use the ocean analogy: we might be seeing only the surface of the universe.
7. End with possibility: curiosity and the fire analogy.

Tone reminders: Keep the writing personal, curious and slightly opinionated. Also avoid using em dashes in the text.

Response 1

The first output devoted roughly 70% of its length to explaining what dark matter is (including a full discussion of Modified Newtonian Dynamics), with only about 20–30% directly addressing why dark matter matters. The MOND section was historically interesting but shifted attention away from the central question. The output also contained em dashes despite the prompt asking to avoid them.

Prompt 2 (ChatGPT 5.2 Instant)

Rewrite the previous response so that the focus shifts away from explaining what dark matter is and instead concentrates more strongly on why dark matter matters. The aim is to reduce general descriptive explanation and strengthen the argumentative and interpretive elements of the text. Remove the discussion of Modified Newtonian Dynamics entirely, as it is not necessary for this stage of the argument. Use the freed space to deepen the central claim about the importance and implications of dark matter.

Add a section that reflects on the state of fundamental physics after the discovery of the Higgs boson. Emphasize that while the Standard Model is considered complete in the sense that it successfully

explains all observed particle physics experiments, it still fails to account for dark matter, which constitutes a much larger fraction of matter in the universe than ordinary baryonic matter. Expand on the idea that dark matter may not simply be isolated particles, but could possess its own internal structure, organizing principles, or symmetries. Ensure that these ideas connect naturally to the concluding reflection on curiosity. Maintain a confident, reflective, opinionated tone suitable for a Substack style essay. Also avoid using em dashes in the text.

Response 2

The second output overcorrected. Only around 20% addressed what dark matter is, while about 80% focused on why it matters. References to the Standard Model were repeated too frequently, and the curiosity-driven tone from the first response was noticeably toned down.

Prompt 3 (ChatGPT 5.2 Instant)

In the previous prompt, the response shifted away from discussing the history and basic nature of dark matter and focused much more heavily on the specific aspects I asked about. While this change was directionally what I wanted, it went too far. In the first response, roughly 70% of the content explained what dark matter is, with about 30% devoted to why it matters. In the second response, this balance was almost reversed. What I want instead is a more balanced approach, approximately 40% explaining what dark matter is, including sufficient historical and conceptual background, and 60% explaining why it matters. Additionally, references to the Standard Model were repeated too frequently in the previous response. The idea itself is important and should remain, but it needs to be stated concisely and without unnecessary repetition. Another issue is that the sense of curiosity and intellectual intrigue was noticeably toned down compared to the first response. I want that curiosity driven tone restored to the same level as before.

Response 3

This response achieved the requested 40/60 balance. The argument is clear and well structured, showing that dark matter challenges how complete our current understanding of physics really is. This became the final version used in the report (see Appendix A.1).

B.2 Task 2: Prompts and Supplementary Material

Original Prompt (Unsuccessful)

I am going to give you a text that I have written for a newspaper article.
 The target style is shock journalism (tabloid-style science reporting).
 The core ideas and structure are already present in the text. Your role is to refine and elevate it so that it fully embodies the required style.
 You may: rearrange ideas for better flow; adjust structure for pacing and impact; refine wording to sharpen tone and intensity; emphasise big-picture ideas rather than technical detail.
 Do not get pulled into overly technical or minor details from the research paper. Focus on the dramatic, big-picture narrative.

Style Targets (Must Follow)

- 1. Paragraph Pacing:** Keep paragraphs short (1–3 sentences maximum). Frequently use single-sentence paragraphs. Alternate punchy lines with slightly longer explanatory lines. Repeat the main hook once later in the article using different wording. Vary paragraph length dramatically. Do not space ideas evenly. Let certain thoughts trail off for effect.
- 2. Vocabulary & Tone:** Use vivid, dynamic verbs (e.g., barrelling, blasting, snapping, peering, shedding light). Use emotional adjectives sparingly but powerfully (e.g., stunning, terrifying, colossal, elusive). Include no more than two human-scale analogies. Use a modern internet-style cadence and sentence rhythm, but do not force slang. Sound genuinely enthusiastic, not exaggerated or performative.

[Draft text was appended below; see Revised Prompt for the full draft.]

This prompt produced the article shown in Appendix A.2. While the output initially seemed effective, group feedback identified the tone as literary rather than journalistic. Lines such as “A blur. A scatter. A nightmare to measure” read more like spoken-word poetry than tabloid science reporting. The prompt’s competing instructions caused the AI to prioritise pacing rules over the newspaper format.

Revised Prompt

I am going to give you a text I have written about the S1 dark matter stream (the so-called “dark matter hurricane”). Your job is to turn it into a 500–700 word tabloid newspaper article for a general audience.

Primary instruction – this overrides everything else: The output must read as a real newspaper article. That means it must have:

- A strong opening lead (1–2 sentences that hook the reader)
- A nut graf early on that tells the reader what this story is actually about
- Short paragraphs (1–3 sentences), as is standard in tabloid journalism
- At least one attributed quote or paraphrased expert statement
- A clear emotional arc: alarm → explanation → reassurance → excitement
- A closing paragraph that leaves the reader with a strong final thought

Secondary instruction: Use vivid, dynamic verbs. Keep the tone genuinely enthusiastic, not exaggerated or performative. Use no more than two human-scale analogies. Avoid technical jargon. Do not write in a literary or poetic style. Do not use isolated single-word sentences or sentence fragments as standalone paragraphs. The output should feel like something published on Express.co.uk or the Daily Mail science section, not like a prose poem.

Draft text:

Scientists are saying that a hurricane of invisible matter is screaming through us right now at speeds reaching up to 1,800,000 km/h. By invisible matter, I mean dark matter, which basically is matter that doesn't interact with anything. If it doesn't interact with anything (so far), then how do we know it exists? The answer is gravity. It interacts through gravity.

Essentially, scientists today know that wherever there is matter, in this case a galaxy, a large halo of dark matter surrounds it. The culprit of this dark matter screaming at insane speeds is S1. Meet S1: a stream of stars moving in the neighbourhood of the Milky Way galaxy as a pack. Essentially, this is a signature of something that happened about 9 billion years ago.

Back then, a small galaxy compared to our beloved Milky Way, about 100 times smaller in mass, smashed into the Milky Way. We are essentially at a crime scene, trying to gather evidence of something that happened 9 billion years ago. Now the question is: is this hurricane a problem for humanity? Should we build shelters or something to protect ourselves from this massive storm? The answer is no, we don't need to, because dark matter doesn't interact directly with anything. In fact, this hurricane is very helpful for scientists.

The thing is, because of the nature of dark matter, it is really hard to detect. What makes it even harder is that dark matter is chaotic. Think of it like many toddlers moving around in a field. Every child is moving in their own direction. Essentially, it is a mess. But since we know the stream of S1 particles is moving in a particular direction, the dark matter associated with it has to be moving in a particular direction as well.

Think of the dark matter associated with S1 like soldiers marching in one particular direction. What this does is help us in the detection of dark matter. Since we know the direction of S1, we can adjust our detectors so that the probability of detecting dark matter increases. There are many theories about what this dark matter could actually be, and each has its own methods of indirect detection. We have our fingers crossed that hopefully one of these methods will allow us to detect dark matter. The problem is that it might be that our technology is still not advanced enough to detect it, or it might be that dark matter behaves in ways we don't yet understand.

Anyway, this is one of the best bets we have for dark matter detection, and the storm is very much worth watching.

ChatGPT Output (Revised Prompt)**ChatGPT 5.2 Instant**

An invisible “hurricane” is tearing through Earth right now at a staggering 1.8 million kilometres per hour, and scientists say we are right in the middle of it.

Before you panic, this is no ordinary storm. It is made of dark matter, the mysterious substance that does not collide with our bodies, our buildings, or even the air we breathe.

Researchers call this cosmic blast the S1 stream, a vast flow of stars and dark matter sweeping through our region of the Milky Way. It is the lingering aftermath of a violent galactic crash that happened around nine billion years ago, when a much smaller galaxy smashed into ours.

What we are experiencing today is essentially the debris from that ancient collision, still racing through space, and through us.

“Dark matter does not interact with ordinary matter in the way we are used to,” scientists explain.

“It passes through everything almost completely unnoticed, except for its gravitational pull.”

That gravity is the key reason we know it exists at all.

Astronomers have long observed that galaxies behave as if they are surrounded by far more mass than we can see. Stars move too fast to be held together by visible matter alone. The missing ingredient is dark matter, forming an invisible halo around galaxies like our own.

The S1 stream is a rare and valuable clue in this otherwise chaotic picture.

Normally, dark matter is thought to move in all directions, more like a swarm than a structured flow. But S1 is different. It is a coherent stream, meaning its particles are all moving in roughly the same direction.

That makes it incredibly important.

“It is like the difference between a crowd wandering randomly and a group marching in formation,” researchers say. “A stream gives us something we can actually aim for.”

For scientists trying to detect dark matter, one of the biggest unsolved problems in physics, this is a breakthrough opportunity.

Experiments deep underground are designed to catch the faintest possible interactions between dark matter and ordinary particles. The problem is, those interactions are extremely rare and unpredictable.

But with S1, scientists have a target.

Because the stream has a known direction and speed, detectors can be tuned to look for signals coming specifically from that flow. That could significantly increase the chances of finally spotting dark matter directly for the first time.

So, is this “hurricane” dangerous? Not at all.

Despite its dramatic name and incredible speed, the S1 stream poses no threat to life on Earth. Dark matter simply does not interact strongly enough with normal matter to cause harm. It passes through us constantly, silently, and harmlessly.

In fact, this cosmic storm could be one of the best things to happen to dark matter research in years. Scientists still do not know what dark matter is made of. There are multiple competing theories, each predicting different ways it might be detected. It could be that current technology is not yet sensitive enough, or that dark matter behaves in ways we do not fully understand.

But S1 offers a rare advantage: structure in the chaos.

And in a field where signals are faint and elusive, that structure could make all the difference.

For now, the invisible hurricane continues to race through our planet, unnoticed by everyone except the scientists watching closely.

And if they are right, this ancient galactic storm could finally help reveal one of the universe’s deepest secrets.

Claude Output (Revised Prompt)**Claude Sonnet 4.6****Dark matter 'HURRICANE' blasting through Earth at 1.8 million km/h but scientists say don't panic**

A ghostly hurricane of invisible matter is barrelling through your body right now at a jaw-dropping 1.8 million kilometres per hour and you cannot feel a thing.

Astronomers call it the dark matter hurricane, and they say it is the wreckage of a cosmic crime scene from 9 billion years ago that could finally help crack one of the biggest mysteries in physics. The storm is driven by something known as the S1 stream, a pack of stars moving together through our corner of the Milky Way at extraordinary speed. They are not local. They are intruders.

According to the team behind the discovery, those stars are the leftover guts of a small galaxy that smashed head-on into our own around 9 billion years ago. The Milky Way swallowed the attacker whole but the debris is still racing through the neighbourhood today.

"S1 is essentially the fingerprint of an ancient galactic collision," researchers say. "And wherever those stars go, a torrent of dark matter follows."

Dark matter is the universe's biggest open secret. Scientists are convinced it is everywhere, making up most of the matter in the cosmos, but no one has ever caught a single particle of it.

The reason it is so slippery is simple. It does not interact with ordinary stuff. Light passes straight through it. Your hand passes straight through it. The only reason we even know it is there is gravity: galaxies, including ours, are wrapped in enormous invisible halos of the stuff, and that gravity tugs on everything around them.

So should we be running for cover from this 1.8-million-kilometre-an-hour storm? Should we be building shelters? Researchers are clear: absolutely not.

Because dark matter ignores ordinary matter, the hurricane is already passing through every person on Earth, every wall and every continent, without leaving so much as a tickle. You have been sitting inside it your whole life.

In fact, scientists are thrilled about the storm rather than worried by it. It could be the best shot they have ever had of catching dark matter in the act.

Most dark matter swirling around the galaxy is chaotic, like a field full of toddlers running off in every direction at once. That randomness makes it horrendously difficult to pin down.

But the dark matter caught up in S1 is different. Because the S1 stars are all moving together in one direction, the dark matter dragged along with them is moving like soldiers marching in formation: coherent, focused, and predictable.

That is gold dust for physicists. By pointing their detectors in the right direction, they can dramatically boost their odds of finally registering a hit.

Experiments buried deep underground around the world are already racing to do exactly that. There are competing theories about what dark matter actually is, and each one comes with its own detection strategy.

The catch is that our technology may still not be sensitive enough, or dark matter may behave in ways nobody has yet predicted. Either way, the hurricane has handed scientists their clearest signpost so far.

For everyday life, the great invisible storm changes nothing. You will not feel it, see it, or be harmed by it. But somewhere beneath the planet's surface, in shielded laboratories built to listen for the faintest whisper of the cosmos, physicists are watching and waiting.

If they catch it, the wind from a galaxy that died 9 billion years ago will have helped solve one of the greatest puzzles in science.

B.3 Task 3: Prompts and Supplementary Material

Prompt 1 (ChatGPT 5.2 Instant)

Understand the format shown below and use it as the structural template.

Reference Format:

PODCAST: SYMMETRIES AND CONSERVATION LAWS

=====

Speakers: ALEX and JESS

JESS: [Slightly panicked] Okay, you have to help me.

ALEX: [Mock serious] Oh no. Of all the weeks to miss.

Your Task: Using that exact style and structure, convert my podcast into the same format.

Requirements:

1. Title: *The Missed Talk*
2. Person 1: Sam (1st year KCL physics, international student from the US)
3. Person 2: Alex (1st year KCL physics, local London student)
4. Add voice performance directions in square brackets throughout

Important: Do not remove content from my draft. Keep the physics explanations intact. Only restructure and format it to match the reference style.

[Draft text was appended below; see Appendix A.3 for the final formatted version.]

Response 1

The output correctly reformatted the draft into the reference style. Character names were set in capitals, performance directions were added in square brackets throughout, and the physics content was preserved without alteration. No changes were needed. The formatted version became the final podcast script used in the report (see Appendix A.3).

B.4 Task 4: Prompts and Supplementary Material

Prompt 1 (Claude Sonnet 4.6)

Create a short educational comic script for children aged around 12 that explains the idea of wave dark matter in a fun, imaginative, and easy-to-understand way. The goal of the comic is to communicate one central idea: that there may be invisible waves of dark matter all around us, passing through everything all the time, even though we cannot see them directly.

The story should follow a simple narrative. A kid is visiting a science museum with their father when the child notices something about dark matter. Curious, the child begins asking questions about dark matter. This begins a conversation between the kid and the father that gradually introduces the idea that dark matter might behave like waves filling the universe, moving through planets, people, and space itself. At one point in the discussion, the child becomes confused and asks how something can be invisible but still real. To explain this, the father uses a relatable analogy: Wi-Fi. He explains that Wi-Fi signals are also invisible, we cannot see them, but we know they are there because they affect the world around us. For example, when the Wi-Fi suddenly stops working at home, the kid's five-year-old younger brother immediately starts screaming from his room because his tablet stops working. In the same way, scientists cannot see dark matter directly, but they can observe its effects, such as the way galaxies move and stay together. The conversation continues with the father explaining that dark matter might behave like giant waves spread across the universe, somewhat like ripples moving through an invisible cosmic ocean. The tone of the story should be curious, playful, and slightly mysterious, making the invisible idea of dark matter feel real and exciting for a young reader.

Response 1

Claude Sonnet 4.6 — First Draft

THE INVISIBLE OCEAN — A Comic About Dark Matter

Page 1, Panel 1: Museum. Jake (12) and Dad walk under a banner: "THE UNIVERSE – MORE THAN MEETS THE EYE." **Jake:** Dad, look at that display!

Panel 2: Exhibit shows galaxy with dark wisps. Label: "DARK MATTER – 27% OF THE UNIVERSE."

Jake: Twenty-seven percent?! So... where is it?

Panel 3: Dad crouches to Jake's level. **Dad:** It's *everywhere*. Even passing through us right now.

Page 2: Jake asks how something invisible can be real. Dad introduces the Wi-Fi analogy. Thought bubble: Tommy watches his tablet; the router dies; Tommy erupts ("NOOOOOOOO! MY SHOW!!").

Page 3: Jake connects the idea: invisible but real. Dad explains dark matter evidence. Jake asks about waves; Dad introduces the wave idea and compares dark matter to enormous waves stretching across galaxies.

Page 4: Full artistic panel: Earth floating in space surrounded by glowing ocean-like ripples flowing through everything. Dad describes "a silent, invisible ocean."

Page 5: Jake, awestruck, says "Then I want to be one of the people who finds them." Caption: "Dark matter may be all around us... We just haven't figured out how to say hello yet."

Prompt 2 (Claude Sonnet 4.6)

There are a couple of things I think we could improve in the comic. First, the banner that reads “THE UNIVERSE: MORE THAN MEETS THE EYE” is not very compelling. Replace it with something more intriguing that would realistically catch a child’s attention.

Second, where the exhibit says “Dark Matter: 27% of the Universe,” simplify and emphasize that around 85% of the matter in the universe cannot be seen. This communicates the idea more dramatically.

Avoid going into too much detail about galaxies spinning. Keep it simple: the universe only makes sense if there is some invisible substance affecting it.

Finally, the father should naturally introduce the wave idea himself rather than having the child suggest it. This connects more naturally with the Wi-Fi analogy, since Wi-Fi signals are also waves.

Response 2

The revised version incorporated all requested changes. The banner became “WARNING: MOST OF THE UNIVERSE IS MISSING.” The 85% statistic replaced the 27% figure. The galaxy rotation explanation was simplified. The father now introduces waves naturally, connecting them to the Wi-Fi analogy. However, the script was still too long to fit on one A4 page.

Prompt 3 (Claude Sonnet 4.6)

This has a bit too many words. We need to fit it on one A4 page, so let’s cut it down a little. At most, we have space for 10 panels overall.

Response 3

Claude Sonnet 4.6 — Final Script (10 Panels)

THE INVISIBLE OCEAN — A Comic About Dark Matter

Panel 1: Museum. Jake (12) and Dad walk under a banner: “WARNING: 85% OF THE UNIVERSE IS INVISIBLE.” **Jake:** Wait, 85%?! Dad, where is it all?! **Dad:** Scientists call it dark matter. And here’s the wild part, it’s passing through us right now.

Panel 2: Jake stares at his hands. **Jake:** But how can something invisible even be real? **Dad:** Think about Wi-Fi. You’ve never seen a Wi-Fi wave. But you know it’s real. Remember last Tuesday?

Panels 3–5: Thought bubble sequence. Tommy happily watching tablet → Router blinks off, dark screen → Tommy ERUPTS: “NOOOOOOOO! MY SHOW!!!”

Panel 6: Back in museum, Jake laughing. **Jake:** Okay, invisible, but definitely real. **Dad:** Same with dark matter. The universe only makes sense if something invisible is affecting everything around us.

Panel 7: **Jake:** So what *is* it exactly? **Dad:** It might behave like waves, enormous ones, stretching across entire galaxies.

Panel 8: Full artistic panel: Earth wrapped in soft glowing ripples flowing through planet, buildings, people. **Dad:** A silent, invisible ocean filling all of space. Waves rolling through planets, through stars, through *you*, right now.

Panel 9: Close-up of Jake, awestruck. **Jake:** So we’re swimming through it... without even knowing? **Dad:** Maybe. The universe keeps hiding its best secrets.

Panel 10: Wide shot. Jake and Dad tiny beneath galaxy mural with faint wave ripples. **Jake:** Then I

want to be one of the people who finds them. **Caption:** *Dark matter may be passing through you right now. We just haven't figured out how to say hello yet.*

B.5 Task 5: Prompts and Supplementary Material

Prompt 1 (ChatGPT 5.2 Instant)

Can you give me a long list of possible opening hooks or analogies for a three-minute explainer video arguing in favour of the budget cuts? I want to see as many different framings as possible so I can pick the one that lands best with an 18–25 year old audience.

Response 1

ChatGPT returned roughly twenty hook variants. Most were unusable, but one stood out: framing the entire debate as a question of where a single £1 should go. This framing carried through the rest of the script.

Prompt 2 (ChatGPT 5.2 Instant)

Using the £1 hook above and the priority-funding rationale we discussed earlier, can you put together a rough script that argues in favour of the budget cuts? The group's main points are: (a) research budgets are limited and have to be prioritised, (b) tax money should go to research with a clearer chance of paying off, (c) curiosity-driven fields like dark matter are a longer gamble than applied areas like health technology, and (d) focusing on near-term economic impact is the more responsible choice. Aim for roughly three minutes when spoken.

Response 2

ChatGPT 5.2 Instant

Why the UK Is Right to Rethink Its Science Budget

The UK invests several billion pounds in scientific research every year, channelled mostly through UK Research and Innovation. That funding has built one of the most respected research systems in the world, supporting everything from cancer treatments to experiments at the edge of fundamental physics.

But the government has now announced that some of that funding is going to be redirected. Areas like particle physics, astronomy and nuclear physics are facing reductions, while more money is being moved into research linked to economic growth — clean energy, artificial intelligence, life sciences, and advanced manufacturing.

It is easy to see why this feels uncomfortable. Curiosity-driven science has, over the years, produced extraordinary returns. The web came out of physics. Modern medical imaging came out of physics. Whole industries can be traced back to research that, at the time, had no obvious commercial purpose.

So why cut anything at all?

The honest answer is that no government has unlimited money for research. Every pound spent on one programme is a pound that cannot be spent on another. When budgets are stretched, trying to support every field equally tends to leave none of them properly resourced. The government's case is that concentrating funding in areas with clearer near-term payoff gives the country a better chance of staying competitive globally.

There is something to that. Research is not only a search for knowledge; it is also a long-term economic strategy. AI, clean energy and the life sciences are widely expected to dominate the next few decades of growth. Putting weight behind them now is a bet on the industries the UK wants to lead in by the 2040s.

None of this means abandoning curiosity-driven research. It means accepting that not everything can grow at once, and that some areas have to take priority over others when public money is tight. Given rising facility costs and the broader pressure on government spending, focusing on research that can deliver tangible results in a reasonable timeframe is a defensible position rather than a reckless one.

The question is not whether to fund discovery science. It is whether every part of it has to be funded at the same level, at the same time, regardless of what else the country needs.

Too formal and too neutral. The £1 hook was not used in the opening, and the counterargument was conceded too early.

Prompt 3 (ChatGPT 5.2 Instant)

This reads more like a balanced opinion essay than an opinionated short-form video. Rewrite it so the £1 hook opens the piece, drop the World Wide Web and MRI examples since they weaken the position before it has been properly stated, and tighten the tone so it sounds like something a young viewer would actually sit through.

Response 3

Improved. The £1 hook now opened the script and the concession was gone, but several sentences ran 30–40 words long and would not work when spoken aloud. Some claims were also too absolute, in particular the statement that dark matter research was funded to the same extent as health technology.

Prompt 4 (ChatGPT 5.2 Instant)

Break the long sentences up so they work when spoken. Soften the absolute claims with qualifiers where needed, and replace policy jargon like “budget allocation” with simpler phrasing. Keep the gambling analogy but tighten the punchline.

Response 4

Close to usable. From this point the script was taken in hand by the group, loosened further during recording to suit spoken delivery, and finalised as the version reproduced in Appendix A.5.

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